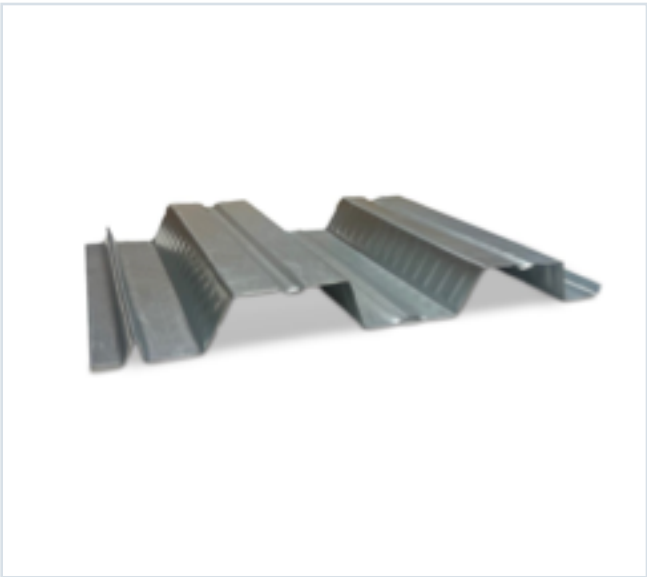


DECKING PRODUCT SUBMITTAL

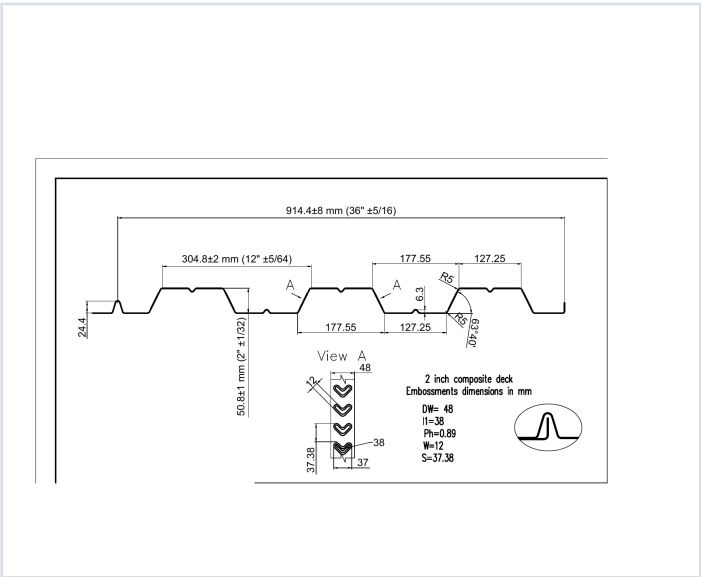
2CD-FY50-22-G90

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	22 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 22 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0295	1.6	50	0.247	0.254	7407	7595	0.311	0.290

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

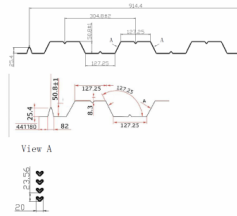
2CD-FY50-22-G90 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E ksi	S _x (inch ³ per foot)	S _y (inch ³ per foot)	ASD (E _x + 16E _y)		I _x (inch ⁴ per ft.	I _y (inch ⁴ per ft.
						M _x / ft. per ft.	M _y / ft. per ft.		
22	0.0295	16	50	0.247	0.254	7407	7595	0.381	0.290
20	0.0358	19	50	0.329	0.336	9860	10000	0.393	0.373
18	0.0474	2.5	50	0.483	0.500	14760	14960	0.548	0.530
16	0.0598	3.2	50	0.645	0.644	19321	19271	0.703	0.683

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _v /V _u) (ksi/ft)		Web Crippling (R _v /V _u) (ksi/ft)	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	627
20	278	290	297	340	398
18	245	245	245	245	245
16	460	507	507	507	507

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	137	137	137	137	137	137	137	137	137	137
	20	181	181	181	181	181	181	181	181	181	181	181
	18	273	273	273	273	273	273	273	273	273	273	273
	16	358	358	358	358	358	358	358	358	358	358	358
Double	22	131	131	131	131	131	131	131	131	131	131	131
	20	196	196	196	196	196	196	196	196	196	196	196
	18	277	277	277	277	277	277	277	277	277	277	277
	16	357	357	357	357	357	357	357	357	357	357	357
Triple	22	176	176	176	176	176	176	176	176	176	176	176
	20	253	253	253	253	253	253	253	253	253	253	253
	18	346	346	346	346	346	346	346	346	346	346	346
	16	446	446	446	446	446	446	446	446	446	446	446

Note: 1. All section properties and ASD (E_x + 16E_y) uniform loads are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span lengths include center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for flexural stress limitations are: Single and Two Span: $M = \frac{wL^2}{8}$; Three Span: $M = \frac{wL^2}{10}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	13	13	13	13	13	13	13	13	13
	20	13	13	13	13	13	13	13	13	13	13	13
	18	13	13	13	13	13	13	13	13	13	13	13
	16	13	13	13	13	13	13	13	13	13	13	13
Double	22	20	20	20	20	20	20	20	20	20	20	20
	20	20	20	20	20	20	20	20	20	20	20	20
	18	20	20	20	20	20	20	20	20	20	20	20
	16	20	20	20	20	20	20	20	20	20	20	20
Triple	22	24	24	24	24	24	24	24	24	24	24	24
	20	24	24	24	24	24	24	24	24	24	24	24
	18	24	24	24	24	24	24	24	24	24	24	24
	16	24	24	24	24	24	24	24	24	24	24	24

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-22-G90 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-8"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-8"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-3"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-8"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-8"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-8"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-8"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	397	384
5	39	400	400	400	400	400	370	337
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	290	280	272	257	243	230	220	210
4.5	296	298	290	280	276	260	247	236
5	304	295	290	277	268	259	249	240
5.5	304	300	291	285	272	262	254	246
6	307	307	304	304	298	290	284	276
6.5	307	307	307	304	304	298	290	284

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	260	256	250	236	229	214	201	190
4.5	267	267	261	246	238	226	214	203
5	277	277	271	254	246	234	221	210
5.5	284	284	278	260	251	239	226	215
6	290	290	284	266	257	245	232	221
6.5	290	290	284	266	257	245	232	221